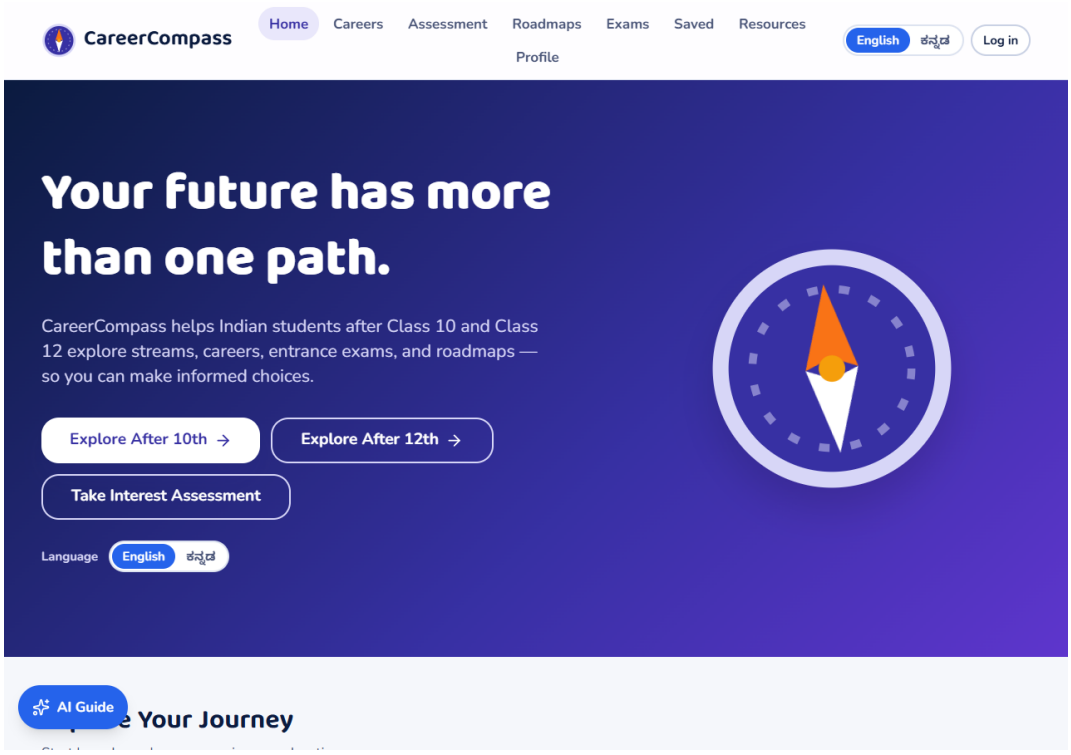
CareerGuide

CareerCompass: project guide and end-to-end workflow

Prepared 30 October 2026 • Plain-language handover

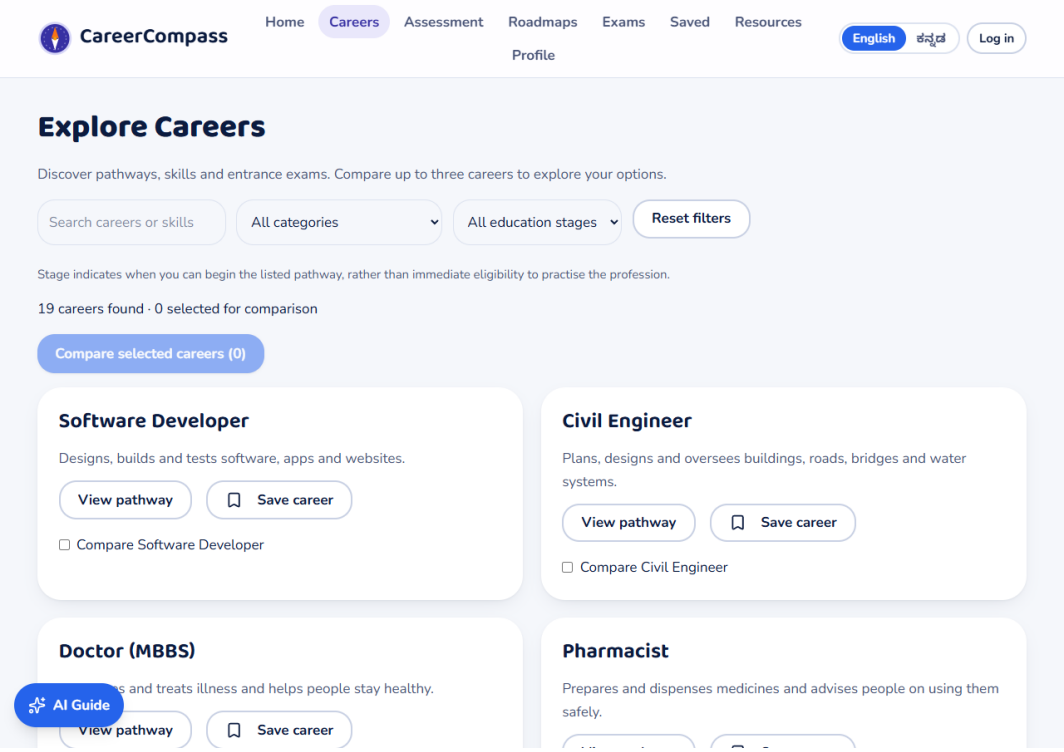
- **Purpose:** help Indian students explore study choices after Class 10 and Class 12, discover careers and plan their next steps.
- **What changed:** the original Lovable foundation now has a connected career directory, interest assessment, visual roadmaps, downloadable PDFs, saved careers, account backup and an AI companion.
- **Identity:** the website is CareerCompass. Its original Lovable project name is Bright Paths India; this is why confirmation emails use that name.
- **Where it runs:** the updated website currently runs locally at http://127.0.0.1:5174. A GitHub update saves source code; it does not by itself confirm a successful production deployment.



Actual local website: home page.

1. What a student can do

- **Home:** start with After 10th or After 12th guidance. These options remain on the home page instead of taking extra navigation space.
- **Careers:** search 19 career pathways, filter by category and education stage, save a career directly from its card and compare 2-3 careers by work, skills and pathway.
- **Assessment:** answer eight interest questions. Matching interest tags produce up to six suggestions. This is a simple guidance tool, not an AI prediction or a validated psychological assessment.
- **Roadmaps:** search with live suggestions (including UI/UX), use Search or Enter, pick a career, tick completed milestones and download an illustrated colour PDF. Two example pathways are shown.
- **Saved and history:** bookmarks and up to 20 assessment results stay in the current browser.
- **Resources:** links to SWAYAM, Bharat Skills and National Career Service, plus a course-selection checklist. Contact and About were removed at your request.
- **Exams:** 21 existing exam records with search, stage filters and official links. Later visual and AI changes did not intentionally change the exam feature.



Actual local website: career discovery and direct save controls.

2. The backend: what is connected to what?

Component	Role in the project
React + TypeScript	The pages and interactive controls students see.
TanStack Start + Vite	Routing, local development, production builds and server functions.
Supabase Auth	Account signup, email confirmation, login, password recovery and sessions.
Supabase PostgreSQL	Cloud tables for profiles and account-plan backups.
Groq: GPT-OSS 120B	Generates career-guidance text replies.
Groq: Whisper Turbo	Converts recorded microphone audio into question text.
Browser storage	Stores bookmarks, assessment history, language and roadmap progress locally.
ReportLab / Python	Generates the 19 illustrated downloadable roadmap PDF files.
Lovable / GitHub	Original project environment and versioned source-code repository.

Simple system flow

- **Student browser → website:** pages show the career dataset and collect questions or planning actions.
- **Website → Supabase:** authenticate the student and read/write only the permitted account data.
- **Website server → Groq → website:** the server sends career context and recent conversation; Groq returns a reply.
- **Microphone → server → Groq transcription → text box:** the student reviews the text before sending the question.
- **Roadmap → PDF link:** download an existing file from public/roadmaps. The chat does not generate custom attachments.

3. Account and AI workflows

Signup and login

- 1. A student enters their name, email, a new CareerCompass password, study stage and language, with the required acknowledgement.
- 2. Supabase creates the account and sends a confirmation email. The normal email link verifies ownership; we do not bypass authentication tables.
- 3. After confirming, the student logs in with the CareerCompass password, not automatically their Gmail password.
- 4. Supabase issues a session. The app uses its token for protected pages and server requests.
- 5. Profile shows the student name, language and stage. Password recovery sends a recovery link and provides a new-password page. Delivery and redirect settings must still be checked for the final production host.

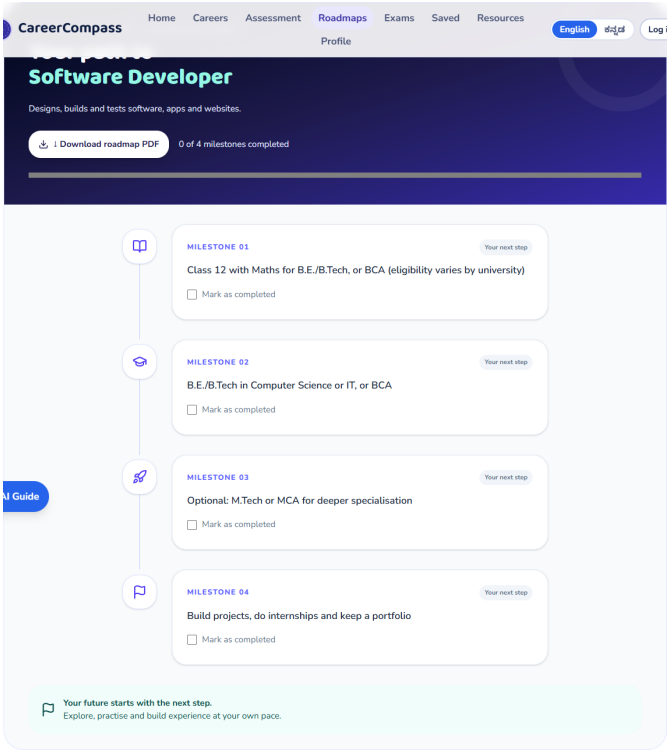
Text conversation

- 1. The signed-in student writes a question. The request goes to an authenticated server function.
- 2. Input validation limits question length and recent history. Longer assistant answers have a separate allowance so a long reply does not break the next turn.
- 3. The server reads GROQ_API_KEY privately and sends the question, recent turns and draft career context to Groq. The configured model is openai/gpt-oss-120b.
- 4. The reply appears in chat. Provider limits and network failures produce readable messages. Questions are sent to Groq; conversation history is held in page memory and clears on refresh.
- 5. The assistant explains options and next steps. The quiz creates an interest shortlist; these are different roles. PDF requests return a polite limitation and point to Roadmaps without an extra Groq call.

Voice conversation

- Record a short question (up to 45 seconds), then stop. The microphone tracks are released.
- Audio goes through the authenticated server to whisper-large-v3-turbo. The student reviews the transcript before asking the guide.
- A spoken question plus an AI answer usually uses two Groq requests. Read reply aloud uses browser speech synthesis rather than another Groq request. Microphone permissions and available device voices still depend on the browser.

4. Roadmaps, saving and account backup



Actual local website: milestone journey with progress and PDF download.

- **Local first:** Save career writes a career slug into browser storage. Roadmap checkboxes store completed pathway steps. Assessment results store a bounded recent history.
- **Cloud backup:** Profile has Save plan to account and Restore account plan. Saving replaces the previous cloud backup; restoring replaces this browser's plan. This is explicit backup/restore, not automatic real-time merging.
- **Tables:** profiles holds account profile information. student_plans stores bookmarks, progress, history and update time, linked to the account user ID.
- **Access:** row-level security on student_plans permits authenticated users to access only their own user_id. The table and policy were checked; live two-account isolation testing remains pending.
- **Downloads:** scripts/generate_roadmaps.py produces the static PDF collection. Regenerate PDFs after changing career pathways so downloads stay consistent.

5. Work completed in phases

- **Phase 1 — connected journeys:** career cards and details, bookmarking, related exam links, nested route fixes and assessment data adaptation.
- **Phase 2 — discovery:** stage/category/search filtering, career comparison and three sourced vocational options: Fitter, Electrician - Power Distribution and COPA.
- **Phase 3 — planning:** assessment history, career checklists, progress and later explicit account backup/restore.
- **Phase 4 — usability:** recovery pages, revised navigation, bottom-left assistant entry, removed Contact/About and useful Resources.
- **Phase 5 — assistance:** authenticated Groq text chat, server-side key, 120B model connection test, voice transcription, read-aloud and polite PDF limitation.
- **Visual refinements:** live search suggestions, search/compare buttons, milestone cards, illustrations and colour PDF roadmap designs.

What was verified - and what is still pending

- Recent automated baseline: 22 tests passed and TypeScript passed. Checks cover content references, filtering, comparison, planning storage, recovery requests, mocked AI handling and conversation-history validation.
- Production builds passed during development. Groq 120B returned a real test reply. Signup confirmation email was found in Gmail and its normal verification link succeeded for the requested account.
- All 19 roadmap PDFs rendered; a sample was visually reviewed. Static download delivery was checked.
- Not fully verified: real microphone transcription on all devices, two-account backup isolation, full recovery lifecycle, native Kannada review and complete mobile/accessibility coverage.
- Lint cleanup is incomplete: formatting was improved, but explicit-any and development-refresh warnings remain. Do not describe the project as having a clean lint result.
- Career guidance remains draft content except where specific official references are supplied. Do not treat admissions advice as guaranteed or current without checking the institution.

6. File map and handover

- **src/routes:** page screens, including login, signup, careers, assessment, roadmaps and AI guide.
- **src/data:** career/exam/quiz content and the three additional vocational careers.
- **src/lib/ai-guide.ts:** authenticated Groq chat request, prompt context and error handling.
- **src/lib/voice.ts:** authenticated audio-transcription request.
- **src/integrations/supabase:** backend clients, session attachment and authentication middleware.
- **src/components/site:** navigation, save buttons, planning, account sync, voice controls and visual roadmap components.
- **supabase/migrations:** student-plan table and owner-only policy.
- **public/roadmaps:** 19 downloadable PDFs. **scripts/generate_roadmaps.py:** PDF generation source.
- **src/test:** automated verification. **.env:** private settings; never upload it. **.env.example:** blank key placeholders.

Run, test and publish

- Install dependencies with npm install. Use npm run dev to run locally; the current local address uses port 5174.
- Use npm test, npx tsc --noEmit and npm run build before release.
- Keep GROQ_API_KEY server-side. The model setting is GROQ_MODEL. Supabase browser settings use its public key; database service credentials must not be shipped to browsers.
- GitHub should receive source, migration files and PDF assets, never private environment values, session tokens, node_modules or generated build output.
- The requested GitHub repository is naveenkumarbuilds-stack/career-compass. Update it with a new normal commit based on its existing main branch. Never force-push or rewrite the Lovable-connected history.
- Publishing still needs a host with server-function support, configured secrets and correct authentication callback URLs. GitHub source and a local server are not the same as a verified public deployment.

Official learning resources

- SWAYAM: https://swayam.gov.in/
- Bharat Skills: https://bharatskill.gov.in/
- National Career Service: https://www.ncs.gov.in/